

Supplementary Information for “Dynamic
Functional Graph Learning for
Subject-Independent Consumer Engagement
Decoding from EEG and Eye Tracking: A
Leakage-Aware NeuMa Study”

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This document (Additional file 1) collects all supplementary figures and tables referenced in the main manuscript. Each caption states, in parentheses, the section of the main manuscript from which the item is referenced, so that every table and figure can be read together with the corresponding part of the main text. Figures and tables are numbered with an “S” prefix; section numbers (e.g., Section 3.7) refer to the main manuscript.

Supplementary Figures

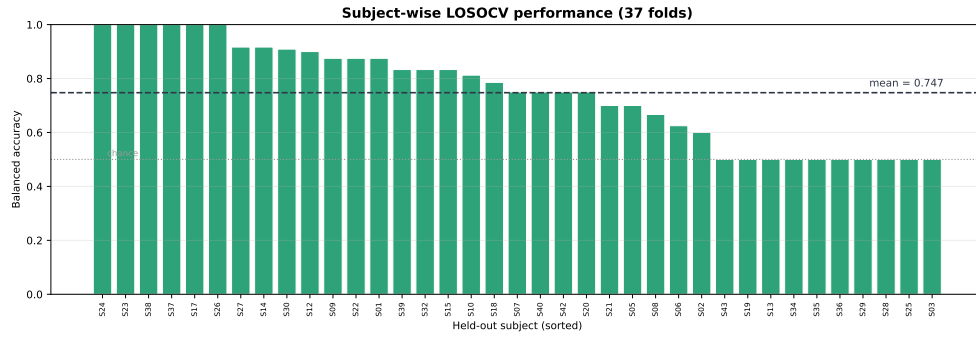


Fig. S1 Subject-wise balanced accuracy under LOSOCV (Section 3.8 of the main manuscript), sorted, with the cohort mean (0.75) and the chance level marked.

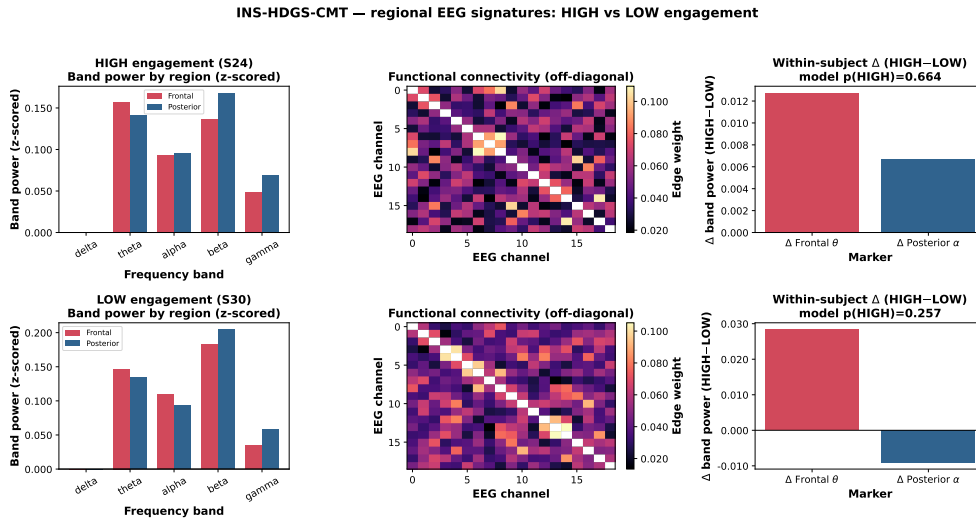


Fig. S2 Regional EEG signatures of the case-study exemplars (S24 HIGH, S30 LOW; Section 3.12): band power by region, off-diagonal connectivity matrix, and the within-subject HIGH-LOW difference for frontal theta and posterior alpha.

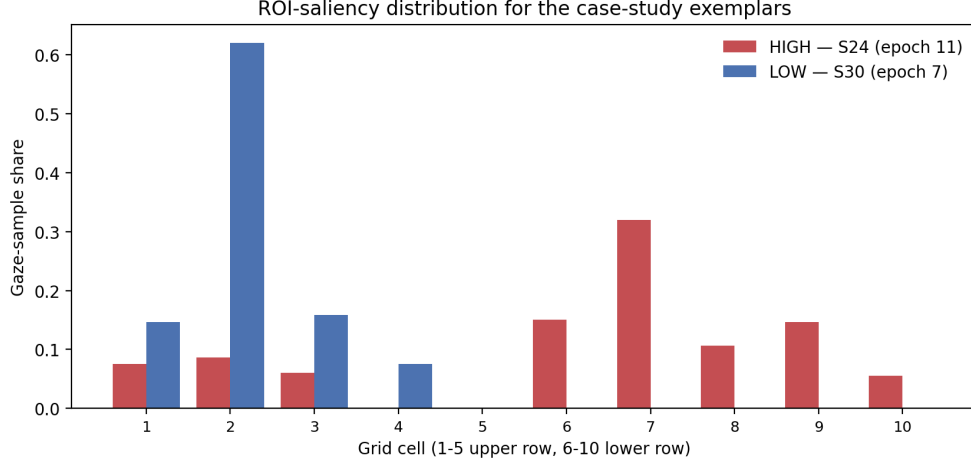


Fig. S3 ROI saliency vector of the two exemplars (Section 3.12): share of gaze samples in each of the ten 5×2 grid cells (1–5 upper row, 6–10 lower row). The HIGH exemplar spans eight cells; the LOW exemplar is confined to four cells of the upper row.

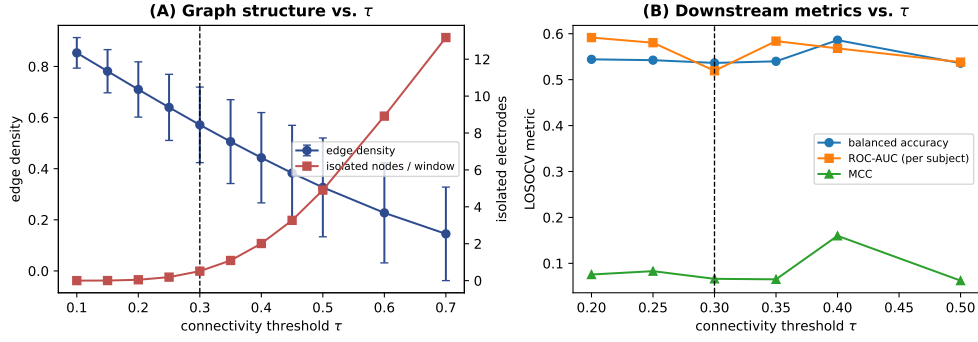


Fig. S4 Connectivity-threshold sensitivity (Section 3.10). (A) Edge density and number of electrodes with fewer than three edges per window against τ in Eq. (4), over all 3,850 windows; dashed line at the value used ($\tau = 0.30$). (B) LOSOCV metrics of the gaze-free branch retrained at each τ (Table S7).

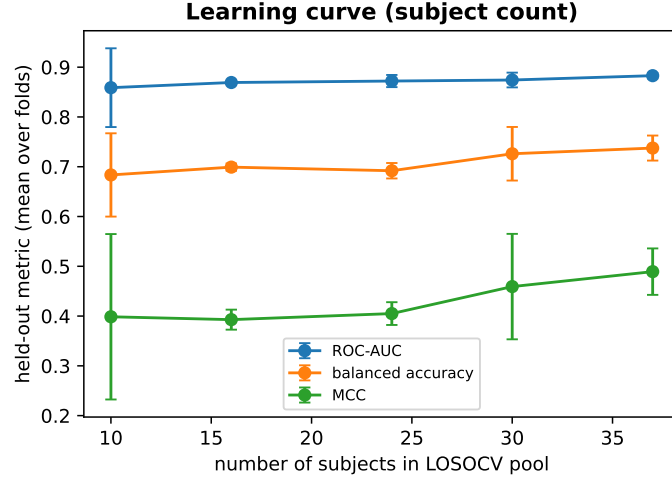


Fig. S5 Subject-count learning curve (Sections 2.6 and 3.10): LOSOCV on random subsets of 10 to 37 participants (three draws each; three-member ensembles, 150 epochs); mean $\pm$ SD over draws.

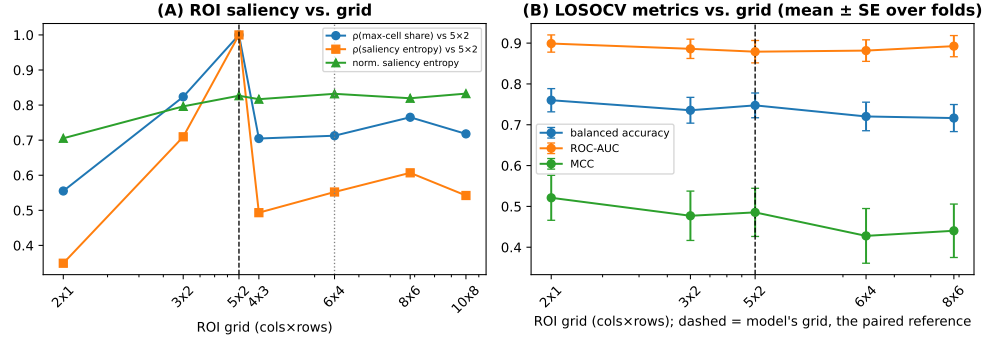


Fig. S6 ROI-grid sensitivity (Section 3.10). (A) Properties of the ROI saliency vector on grids from 2×1 to 10×8 cells and their rank correlation with the 5×2 vector (dashed; dotted: the 24-cell layout of Fig. 1). (B) Full model retrained with each grid (mean $\pm$ SE over 37 folds), with the 5×2 run of Table S11 at the dashed line.

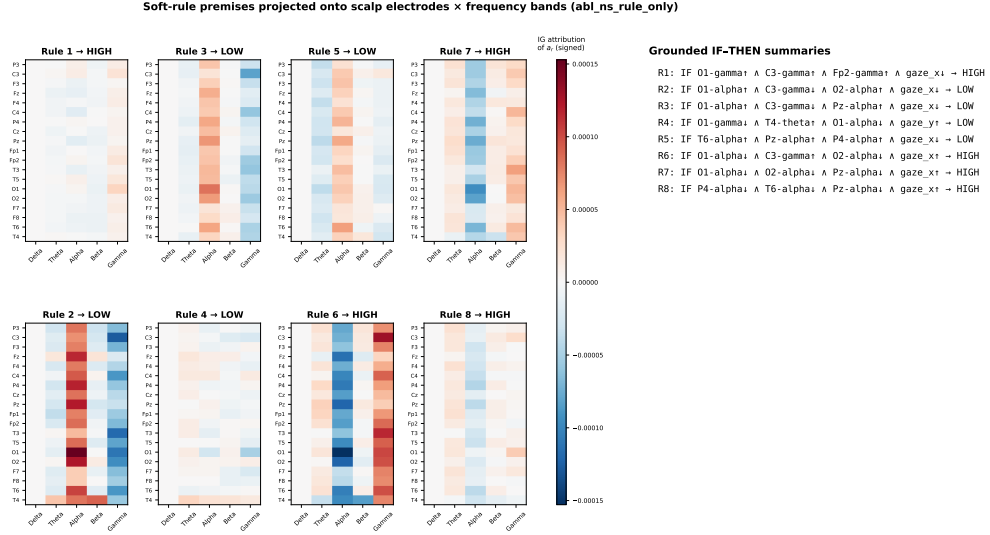


Fig. S7 Attribution-grounded rule premises (Sections 2.5.8 and 3.11.3): integrated-gradient attribution of each rule’s activation to the 19×5 electrode-by-band features on the held-out epochs of the rule-gate-closed model (Table S13); red increases, blue decreases the activation. Right: three largest terms per rule, dominant eye-tracking stream and rule conclusion (compact form in Fig. 10D).

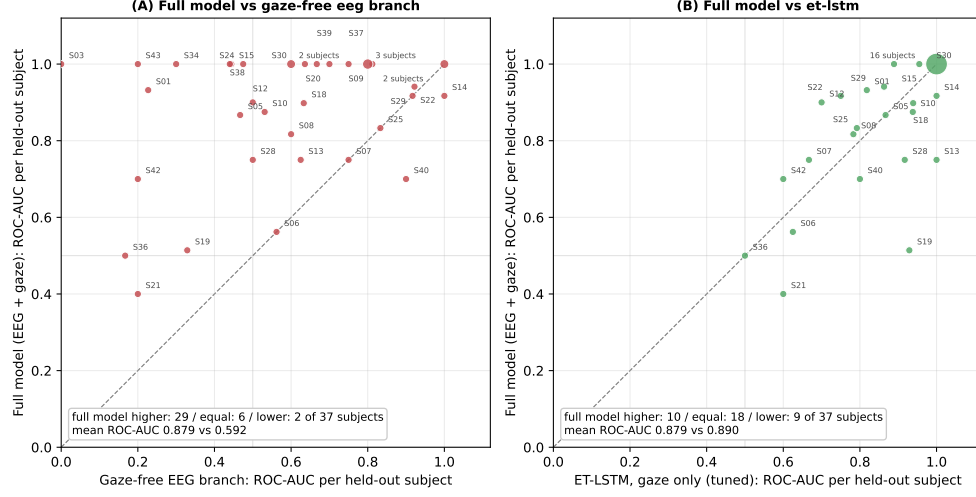


Fig. S8 Per-subject ROC-AUC of the full model against (A) the gaze-free EEG branch and (B) the tuned gaze-only ET-LSTM (Sections 3.2 and 3.5). Points above the diagonal favour the full model. Per-fold ROC-AUC saturates on folds of about nine epochs, so subjects with identical scores are drawn as a single marker, sized by and labelled with the number of subjects it represents; in (B) sixteen subjects share the point (1.00, 1.00). The inset counts subjects above, on and below the diagonal (paired statistics in Table S14).

Supplementary Tables

Table S1 Labelling-regime check (Section 3.10) with the ROI-retained EEG branch of the submitted version under the common budget: the global-threshold regime (37 evaluable subjects) beside the subject-median regime (42 subjects). Not repeated with the gaze-free branch (balanced accuracy 0.55, ROC-AUC 0.59; Table 4).

Labelling regime	Subjects evaluable	Balanced acc.	ROC-AUC
Global threshold (headline regime)	37	0.70 ± 0.16	0.82 ± 0.19
Subject-median (coverage regime)	42	0.69 ± 0.18	0.80 ± 0.18

Table S2 Top integrated-gradients input attributions aggregated across folds (Sections 3.11.1 and 3.11.4; normalised importance).

Modality	Feature	Importance	Interpretation
ROI	ROI saliency vector	0.52	Attended stimulus region
EEG	Posterior alpha power	0.18	Visual attention allocation (alpha suppression)
EEG	Frontal theta power	0.16	Attentional control / engagement
ET	Gaze position	0.13	Overt fixation location
ET	Pupil dynamics	0.01	Arousal / cognitive load
EEG	Frontal functional connectivity	< 0.01	Network integration (see ablation)

Table S3 Connectivity-measure comparison at reduced scale ($N = 16$; Section 3.10). All seven measures are near chance, as is the gaze-free branch at full scale under Pearson correlation (Table 4); the topology nulls of Section 3.4 show that the decision does not depend on the measured connectivity.

Connectivity measure	n folds	Mean subj. AUC	Pooled AUC	Mean MCC
Pearson	15	0.535	0.506	-0.014
PLV (broadband)	15	0.568	0.502	+0.021
PLV (delta)	13	0.596	0.527	+0.141
PLV (theta)	14	0.517	0.457	-0.007
PLV (alpha)	15	0.573	0.527	-0.032
PLV (beta)	15	0.490	0.480	-0.055
PLV (gamma)	15	0.466	0.469	-0.029

Table S4 Mathematical notation of the Methods (Section 2.5), in order of first use.

Symbol	Meaning
$G_t = (V, E_t, A_t)$	Dynamic EEG functional graph at window t
$\rho_{ij}(t)$	Windowed Pearson connectivity between electrodes i, j
PLV_{ij}	Phase-locking value connectivity (alternative measure)
τ	Connectivity threshold sparsifying A_t
α_{ij}	Graph-attention coefficient between nodes i, j
$Z_{\text{EEG} \rightarrow \text{ET}}$	EEG-to-ET cross-attention output
Z_{fusion}	Fused multimodal representation
g	Gated residual fusion weight
a_r	Neuro-symbolic rule activation
R	Aggregated rule evidence
α (bypass gate)	Rule/bypass blend weight (distinct from α_{ij} above)
$L_{\text{div}}, L_{\text{sparse}}$	Rule diversity / sparsity regularisers
L_{mmd}	Subject-embedding alignment penalty
L_{con}	Cross-modal InfoNCE contrastive loss (EEG–gaze embedding alignment; uses no label)
L_{roi}	ROI-attention guidance loss
L_{total}	Combined training objective

Table S5 Composition of the engagement label (Sections 2.4 and 4.6; `engagement_phase3d.py`, which reproduces the held-out labels 347/347). Each term is computed per 5 s epoch, min–max normalised over the pooled epochs of all subjects, weighted and summed; the composite is thresholded at the pooled median (193 HIGH, 192 LOW). EEG terms use the mean of P3, C3, F3 and Fz (named ‘frontal’ by the pipeline); gaze terms use the epoch’s own normalised gaze samples; pupil diameter is not used. Table S15 gives the linear recoverability from each group of terms.

Modality	Term	Weight	Definition
EEG	Theta power	+1.5	band power 4–8 Hz of the P3/C3/F3/Fz mean
EEG	Beta power	+1.2	band power 13–30 Hz
EEG	Alpha power	−1.5	band power 8–13 Hz
EEG	Theta/beta ratio	+1.0	theta power divided by beta power
EEG	‘Asymmetry’	+0.5	mean amplitude of P3 minus mean amplitude of Fz (close to zero for band-passed, average-referenced signals)
Gaze	Horizontal position	+1.0	mean $ x $ over valid samples
Gaze	Vertical position	+1.0	mean $ y $ over valid samples
Gaze	Horizontal dispersion	+1.0	standard deviation of x
Gaze	Stability count	+1.0	number of consecutive samples with $ \Delta x < 0.05$
Gaze	Gaze entropy	−1.0	Shannon entropy of the 20-bin histogram of x

Table S6 Gaze-free EEG branch versus each tuned EEG baseline on MCC and ROC-AUC (37 folds; paired Wilcoxon, zero differences discarded, Holm-corrected over the eight baselines; r_{rb} positive favours the branch; Section 3.6). Complements Table 9. No comparison is significant.

Baseline	MCC			ROC-AUC		
	Median Δ	p_{Holm}	r_{rb}	Median Δ	p_{Holm}	r_{rb}
EEGNet	+0.091	1.00	+0.28	+0.125	1.00	+0.26
GAT	+0.140	1.00	+0.23	+0.000	0.44	+0.38
EEG Transformer	+0.000	1.00	+0.20	+0.000	1.00	+0.01
CNN-LSTM	+0.000	1.00	+0.09	+0.000	1.00	+0.06
TSception	+0.091	1.00	+0.15	+0.000	1.00	+0.12
CNN-BiLSTM	+0.000	1.00	-0.09	+0.000	1.00	-0.13
DeepConvNet	-0.091	1.00	-0.19	+0.000	1.00	-0.04
ShallowConvNet	-0.125	1.00	-0.14	+0.000	1.00	-0.09

Table S7 Connectivity-threshold sensitivity (Sections 2.5.2 and 3.10). Graph structure pooled over the 3,850 windows (density = fraction of the 171 pairs retained by Eq. (4); $\bar{d}$ = mean degree; n_{iso} = electrodes with fewer than three edges per window; f_{iso} = fraction of windows with one). Downstream: gaze-free branch retrained at each τ (three-member ensembles, 120 epochs, 37 folds; mean per-subject ROC-AUC).

τ	Density	$\bar{d}$	n_{iso}	f_{iso}	BalAcc	AUC
0.10	0.85 ± 0.06	15.4	0.00	0.000	—	—
0.15	0.78 ± 0.08	14.1	0.00	0.004	—	—
0.20	0.71 ± 0.11	12.8	0.05	0.044	0.544	0.592
0.25	0.64 ± 0.13	11.5	0.19	0.169	0.542	0.580
0.30 (used)	0.57 ± 0.15	10.3	0.51	0.375	0.536	0.519
0.35	0.51 ± 0.16	9.1	1.09	0.603	0.540	0.584
0.40	0.44 ± 0.18	8.0	2.01	0.776	0.586	0.568
0.45	0.38 ± 0.19	6.9	3.27	0.874	—	—
0.50	0.33 ± 0.19	5.9	4.91	0.930	0.536	0.538
0.60	0.23 ± 0.20	4.1	8.92	0.965	—	—
0.70	0.15 ± 0.18	2.6	13.17	0.976	—	—

Table S8 Baselines under the earlier common fixed budget (Adam, 10^{-3} , weight decay 10^{-4} , batch 32, 100 epochs, no early stopping or search), kept beside the tuned values of Tables 4–6. LOSOCV, 37 folds, uncalibrated operating point, mean $\pm$ SD.

Model	BalAcc	MCC	ROC-AUC	ECE	Brier
<i>EEG encoders</i>					
EEGNet	0.62 $\pm$ 0.19	0.19 $\pm$ 0.34	0.60 $\pm$ 0.26	0.30	0.35
ShallowConvNet	0.62 $\pm$ 0.20	0.15 $\pm$ 0.35	0.66 $\pm$ 0.20	0.30	0.33
CNN-LSTM	0.58 $\pm$ 0.18	0.12 $\pm$ 0.33	0.55 $\pm$ 0.24	0.43	0.44
CNN-BiLSTM	0.56 $\pm$ 0.19	0.02 $\pm$ 0.32	0.58 $\pm$ 0.23	0.37	0.39
EEG Transformer	0.55 $\pm$ 0.16	0.07 $\pm$ 0.30	0.60 $\pm$ 0.25	0.46	0.47
TSception	0.54 $\pm$ 0.18	0.06 $\pm$ 0.34	0.57 $\pm$ 0.25	0.40	0.42
DeepConvNet	0.53 $\pm$ 0.22	0.05 $\pm$ 0.38	0.58 $\pm$ 0.26	0.37	0.41
GAT	0.51 $\pm$ 0.19	-0.03 $\pm$ 0.26	0.49 $\pm$ 0.24	0.08	0.26
<i>Eye-tracking encoders</i>					
ET-GRU	0.78 $\pm$ 0.17	0.47 $\pm$ 0.36	0.83 $\pm$ 0.19	0.06	0.15
ET-LSTM	0.77 $\pm$ 0.19	0.47 $\pm$ 0.39	0.86 $\pm$ 0.19	0.09	0.16
ET-Transformer	0.75 $\pm$ 0.19	0.43 $\pm$ 0.38	0.78 $\pm$ 0.23	0.23	0.24
<i>Multimodal fusion</i>					
Dual Transformer	0.79 $\pm$ 0.16	0.50 $\pm$ 0.35	0.81 $\pm$ 0.23	0.23	0.23
DynamicGAT+ET Transf.	0.78 $\pm$ 0.18	0.46 $\pm$ 0.38	0.78 $\pm$ 0.23	0.22	0.23
Cross-Attention	0.74 $\pm$ 0.20	0.41 $\pm$ 0.39	0.81 $\pm$ 0.20	0.20	0.21
Multimodal Transformer	0.74 $\pm$ 0.19	0.41 $\pm$ 0.39	0.77 $\pm$ 0.22	0.21	0.22
Late Fusion	0.63 $\pm$ 0.21	0.19 $\pm$ 0.39	0.65 $\pm$ 0.23	0.24	0.28

Table S9 Measured cost of the spiking front-end (Section 3.4; `snn_energy_measured.py`): the trained LIF encoder (two layers, 128 units, 10 steps) against a dense twin of the same widths (Linear+GELU+LayerNorm). Latency = mean of 300 single-epoch forward passes after warm-up (CPU: AMD Ryzen 7; GPU: NVIDIA A100). GPU board power rose by no more than 3 W above its 85 W idle for either encoder, so no energy figure is reported; the operation-count projection of the submitted version applies only to neuromorphic hardware and is withdrawn. Trained firing rate 7.5% on the fold-1 checkpoint timed here (10.6% on the fold-1 checkpoint of the submitted 24-node run).

Hardware	Encoder	Batch	Latency (ms)
CPU (Ryzen 7)	spiking (LIF)	1	2.70
	dense twin	1	0.45
	spiking (LIF)	32	3.45
	dense twin	32	0.69
GPU (A100)	spiking (LIF)	1	3.99
	dense twin	1	0.41
	spiking (LIF)	32	4.03
	dense twin	32	0.43

Table S10 Full model versus each eye-tracking and fusion baseline (Section 3.2): paired Wilcoxon over the 37 folds, Holm-corrected over the eighteen tuned baselines within each metric, zero differences discarded; r_{rb} positive favours the full model (ties excluded, so it can be sizeable where many folds tie, as on ROC-AUC). The EEG encoders are compared with the gaze-free branch in Tables 9 and S6. * $p_{Holm} < 0.05$.

Baseline	Balanced accuracy		MCC		ROC-AUC	
	p_H	r_{rb}	p_H	r_{rb}	p_H	r_{rb}
ET-GRU	1.00	-0.01	1.00	-0.11	1.00	+0.18
ET-LSTM	1.00	-0.07	1.00	-0.12	1.00	-0.05
ET-Transformer	1.00	-0.13	1.00	-0.14	1.00	+0.30
Early-Fusion MLP	1.00	+0.12	1.00	+0.07	0.85	+0.29
Late Fusion	1.00	+0.22	1.00	+0.11	0.68	+0.39
Dual Transformer	1.00	-0.04	1.00	-0.13	1.00	+0.24
Cross-Attention	1.00	-0.17	1.00	-0.21	0.85	+0.33
Multimodal Transformer	1.00	+0.02	1.00	-0.04	0.85	+0.35
DynamicGAT+ET Transf.	1.00	-0.07	1.00	-0.17	1.00	+0.21

Table S11 ROI-grid sensitivity (Section 3.10; 385 epochs). Top: the ROI saliency vector on alternative grids (max-cell share, normalised entropy, Spearman ρ with the 5×2 vector). Bottom: full model retrained with each grid, paired against the model's own 5×2 run (BalAcc 0.747, ROC-AUC 0.879, MCC 0.485; results/sensitivity/roi_grid_model_side.csv).

<i>ROI saliency vector (model input)</i>				
Grid	Cells	Max share	Entropy	ρ vs. 5×2
2×1	2	0.74	0.71	0.56
3×2	6	0.51	0.80	0.82
5×2 (model)	10	0.41	0.83	1.00
4×3	12	0.40	0.82	0.71
6×4 (Fig. 1 layout)	24	0.32	0.83	0.71
8×6	48	0.29	0.82	0.77
10×8	80	0.23	0.83	0.72
<i>Full model retrained on each grid (LOSOVC, 37 folds)</i>				
Grid	BalAcc	AUC	MCC	ΔAUC (p)
2×1	0.760 ± 0.173	0.899 ± 0.128	0.521	+0.020 (0.59)
3×2	0.736 ± 0.192	0.886 ± 0.144	0.477	+0.007 (0.72)
5×2 (model)	0.747 ± 0.185	0.879 ± 0.167	0.485	—
6×4	0.720 ± 0.213	0.882 ± 0.160	0.428	+0.003 (0.53)
8×6	0.716 ± 0.202	0.893 ± 0.158	0.440	+0.014 (0.83)

Table S12 Label-free per-subject thresholds for the full model (Section 3.8; 37 folds, mean $\pm$ SD); no strategy uses a test-subject label. First two rows: the transferred threshold and the calibrated operating point of the main text; “transductive” uses all of the held-out subject’s unlabelled probabilities; “causal” uses only the preceding epochs (prevalence-matched quantile at the training prior), falling back to the transferred threshold for the first k epochs. ROC-AUC is threshold-invariant.

Strategy	Type	BalAcc	MCC	Accuracy
Transferred validation-subject threshold (the value tabulated throughout)	fixed	0.747 ± 0.185	0.485 ± 0.359	0.782
Fixed 0.5 (= calibrated operating point, Section 2.6)	fixed	0.719 ± 0.194	0.446 ± 0.378	0.767
Prior quantile, all epochs	transductive	0.740 ± 0.189	0.433 ± 0.339	0.692
Prior quantile, warm-up 2	causal	0.733 ± 0.179	0.429 ± 0.328	0.709
Prior quantile, warm-up 3	causal	0.748 ± 0.183	0.464 ± 0.339	0.735
Prior quantile, warm-up 5	causal	0.746 ± 0.190	0.474 ± 0.355	0.754

Table S13 Fidelity of the rule pathway to the decision (Section 3.11; `rule_fidelity.py` on the fold checkpoints of the full model). α = learned bypass gate of Eq. (19), unchanged from its initialisation (0.570, range 0.567–0.573); “agreement” = share of held-out epochs on which $\arg \max R$ equals the gated decision (39.4%); rule margin versus logit margin $r = -0.69$; rule attention uniform. Post-hoc rows force α at inference (decision at 0.5, no temperature scaling); the last row is trained with $\alpha \equiv 0$ (paired Wilcoxon versus the full model $p = 0.047$, 0.014 and 0.093 for balanced accuracy, ROC-AUC and MCC).

Decision rule	BalAcc	ROC-AUC	MCC
Gated (as reported)	0.719 ± 0.194	0.879 ± 0.167	0.446 ± 0.378
Rule evidence only, post hoc ($\alpha \rightarrow 0$)	0.392 ± 0.185	0.221 ± 0.237	-0.220 ± 0.379
Bypass only, post hoc ($\alpha \rightarrow 1$)	0.725 ± 0.193	0.877 ± 0.167	0.457 ± 0.377
Trained with $\alpha \equiv 0$ (rule-only model)	0.715	0.839	0.438

Table S14 Eye-tracking pathway contribution: full model minus each variant (Section 3.5; 37 fold-matched subjects, uncalibrated operating point; paired Wilcoxon with zero differences discarded, Holm-corrected over the eight pairs tested per metric, r_{rb} and bootstrap 95% CI). Variants: –gaze sequence = sequence branch and fusion transformer removed, ROI vector retained; –ROI = ROI modulation removed; –fusion = cross-modal transformer replaced by single cross-attention; EEG-only = no gaze-derived input; ET-LSTM = tuned gaze-only encoder (Table 5).

Metric	full vs.	Mean Δ [95% CI]	p_H	r_{rb}	W/T/L
ROC-AUC	–gaze seq.	+0.060 [–0.009, +0.138]	0.12	+0.56	17/16/4
ROC-AUC	–ROI	+0.015 [–0.034, +0.070]	1.00	+0.24	9/22/6
ROC-AUC	–fusion	–0.012 [–0.050, +0.020]	1.00	+0.10	9/21/7
ROC-AUC	ET-LSTM	–0.011 [–0.049, +0.023]	1.00	–0.05	10/18/9
ROC-AUC	EEG-only	+0.287 [+0.207, +0.374]	< 0.001*	+0.96	29/6/2
BalAcc	–gaze seq.	+0.051 [+0.009, +0.092]	0.21	+0.52	15/16/6
BalAcc	–ROI	+0.034 [–0.012, +0.076]	0.32	+0.50	13/20/4
BalAcc	–fusion	+0.033 [–0.002, +0.069]	0.32	+0.46	12/21/4
BalAcc	ET-LSTM	–0.009 [–0.073, +0.052]	1.00	–0.01	16/13/8
BalAcc	EEG-only	+0.195 [+0.129, +0.261]	< 0.001*	+0.87	27/5/5
MCC	–gaze seq.	+0.088 [–0.000, +0.172]	0.22	+0.47	15/16/6
MCC	–ROI	+0.059 [–0.027, +0.135]	0.22	+0.56	13/20/4
MCC	–fusion	+0.074 [+0.009, +0.143]	0.22	+0.54	12/21/4
MCC	ET-LSTM	–0.032 [–0.155, +0.088]	1.00	–0.12	12/15/10
MCC	EEG-only	+0.391 [+0.250, +0.535]	< 0.001*	+0.81	27/5/5

Table S15 Linear recoverability of the label from its own terms (Section 3.2; `label_leakage_audit.py`): logistic-regression probe, leave-one-subject-out over the 37 evaluable subjects; pooled and per-fold (mean $\pm$ SD) ROC-AUC and balanced accuracy. The rule-score row is 1.0 by construction; the EEG-5 and gaze-5 rows are the floors for a model that sees only that modality’s defining statistics.

Feature set	Pooled AUC	Fold AUC	Pooled BalAcc	Fold BalAcc
EEG-5 (band power of the P3/C3/F3/Fz mean)	0.605	0.671 ± 0.216	0.571	0.582 ± 0.172
Gaze-5 (gaze statistics)	0.939	0.917 ± 0.137	0.862	0.831 ± 0.187
All ten terms	0.996	0.999 ± 0.004	0.965	0.957 ± 0.103
Rule score (upper bound)	1.000	1.000 ± 0.000	0.989	0.989 ± 0.044

Table S16 Two controls on the same recordings (Sections 3.2 and 3.4). Positive control: resting state (2-min pre-brochure fixation) versus brochure browsing, 5-s epochs, 42 participants, LOSOCV; rungs trained gaze-free with three-member ensembles and no domain-alignment terms unless stated. Behavioural label: whether the viewed product was later bought (Q77), one 5-s epoch per participant–product pair anchored on the product’s longest gaze run and ending 250 ms before the first click on it (2,338 epochs, 418 bought, 42 participants); dwell rows are logistic-regression probes on session-level gaze covariates. Mean $\pm$ SD over folds, uncalibrated operating point.

Model	BalAcc	ROC-AUC	MCC	folds
<i>Positive control: resting state vs browsing</i>				
EEGNet, tuned	—	0.917	—	42
ShallowConvNet, tuned	—	0.944	—	42
Linear encoder on the model’s node features	0.720 ± 0.125	0.850 ± 0.119	0.476	42
+ production DANN/MMD regularisers	0.706 ± 0.117	0.842 ± 0.109	0.449	42
Graph pathway only (plain head)	0.659 ± 0.113	0.856 ± 0.108	0.383	42
Spiking encoder only, raw EEG, 300 LIF steps	0.525 ± 0.096	0.551 ± 0.132	0.070	42
Spiking encoder + graph pathway (plain head)	0.695 ± 0.126	0.809 ± 0.140	0.420	42
Proposed gaze-free branch (as in Table 4: five members, adversarial term retained, no MMD)	0.721 ± 0.121	0.882 ± 0.106	0.484	42
Full model (EEG + gaze, production configuration)	0.887 ± 0.135	0.989 ± 0.016	0.795	42
<i>Behavioural label: product bought vs not bought</i>				
Full model, production configuration	0.566 ± 0.091	0.606 ± 0.118	0.112	42
Linear probe, five EEG label terms (within-epoch)	—	0.51	—	42
Linear probe, five gaze label terms (within-epoch)	—	0.57	—	42
Linear probe, total dwell on the product (session-level)	—	0.85 / 0.88	—	42
Linear probe, dwell + number of runs + views + anchor run	—	0.90 / 0.91	—	42

Table S17 Fold-wise-label robustness (Sections 2.4 and 3.10 of the main manuscript; `scripts/revision/compare_v2.py`). Each run was repeated with the label’s min–max scaler and median fitted on each fold’s training participants only, so that no held-out participant contributes to the construction of its own labels, and with a validation participant drawn by a rule that is identical for the proposed model and every baseline and never selects a single-class participant. All 37 folds and 347 held-out epochs remain evaluable under the fold-wise label. Paired over the 37 held-out participants; Δ = fold-wise minus pooled label, with a 95% bootstrap confidence interval of the mean (10,000 resamples) and the paired two-sided Wilcoxon p (zero differences discarded). For reference, the run-to-run spread of the same configuration over three trainings is 0.030 in balanced accuracy and 0.016 in ROC-AUC (Section 3.4 of the main manuscript).

Run	Metric	Pooled	Fold-wise	Δ [95% CI]	p
Full model	BalAcc	0.747	0.715	−0.033 [−0.092, +0.027]	0.32
	ROC-AUC	0.879	0.872	−0.007 [−0.039, +0.028]	0.47
	MCC	0.485	0.455	−0.030 [−0.140, +0.080]	0.56
Gaze-free EEG branch	BalAcc	0.552	0.550	−0.002 [−0.068, +0.065]	0.88
	ROC-AUC	0.592	0.605	+0.013 [−0.052, +0.078]	0.80
	MCC	0.094	0.091	−0.003 [−0.130, +0.124]	0.99